Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

x	$3x-1$	$3x+1$
–2	–7	–5
–1	–4	–2
0	–1	1
1	2	4
2	5	7

Among the answer choices, the only one not found in the table is 6.

The correct answer is E.

•

For all positive integers m and v, the expression $m \Theta v$ represents the remainder when m is divided by v. What is the value of ((98 Θ 33) Θ 17) − (98 Θ (33 Θ 17))?

A. −10

B. −2

C. 8

D. 13

E. 17

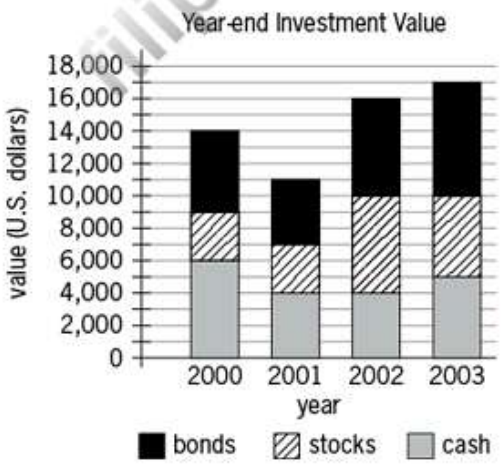
Arithmetic Operations with Integers

First, for ((98 Θ 33) Θ 17), determine 98 Θ 33, which equals 32, since 32 is the remainder when 98 is divided by 33 (98 = 2(33) + 32). Then, determine 32 Θ 17, which equals 15, since 15 is the remainder when 32 is divided by 17 (32 = 1(17) + 15). Thus, ((98 Θ 33) Θ 17) = 15.

Next, for (98 Θ (33 Θ 17)), determine 33 Θ 17, which equals 16, since 16 is the remainder when 33 is divided by 17 (33 = 1(17) + 16). Then, determine 98 Θ 16, which equals 2, since 2 is the remainder when 98 is divided by 16 (98 = 6(16) + 2). Thus, (98 Θ (33 Θ 17)) = 2.

Finally, ((98 Θ 33) Θ 17 − (98 Θ (33 Θ 17)) = 15 − 2 = 13.

The correct answer is D.



•

The chart above shows year-end values for Darnella's investments. For just the stocks, what was the increase in value

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

The chart above shows year-end values for Darnella's investments. For just the stocks, what was the increase in value from year-end 2000 to year-end 2003?

A. $1,000

B. $2,000

C. $3,000

D. $4,000

E. $5,000

Arithmetic Interpretation of Graphs

From the graph, the year-end 2000 value for stocks is 9,000 − 6,000 = 3,000 and the year-end 2003 value for stocks is 10,000 − 5,000 = 5,000. Therefore, for just the stocks, the increase in value from year-end 2000 to year-end 2003 is 5,000 − 3,000 = 2,000.

The correct answer is B.

•

If the sum of the reciprocals of two consecutive odd integers is $\frac{12}{35}$, then the greater of the two integers is

A. 3

B. 5

C. 7

D. 9

E. 11

Arithmetic Operations with Fractions

The sum of the reciprocals of 2 integers, a and b, is $\frac{1}{a}+\frac{1}{b}=\frac{a+b}{ab}$. Therefore, since $\frac{12}{35}$ is the sum of the reciprocals of 2 consecutive odd integers, the integers must be such that their sum is a multiple of 12 and their product is the same multiple of 35 so that the fraction reduces to $\frac{12}{35}$. Considering the simplest case where $a+b=12$ and $ab=35$, it is easy to see that the integers are 5 and 7 since 5 and 7 are the only factors of 35 that are consecutive odd integers. The larger of these is 7.

Algebraically, if a is the greater of the two integers, then $b=a-2$ and

$$\begin{aligned}\frac{a+(a-2)}{a(a-2)} &= \frac{12}{35}\\ \frac{2a-2}{a(a-2)} &= \frac{12}{35}\\ 35(2a-2) &= 12a(a-2)\\ 70a-70 &= 12a^2-24a\\ 0 &= 12a^2-94a+70\\ 0 &= 2(6a-5)(a-7)\end{aligned}$$

Thus, $6a-5=0$, so $a=\frac{5}{6}$, or $a-7=0$, so $a=7$. Since a must be an integer, it follows that $a=7$.

The correct answer is C.

•

What is the sum of the odd integers from 35 to 85, inclusive?

A. 1,560

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. 1,500

C. 1,240

D. 1,120

E. 1,100

Arithmetic Operations on Integers

The odd integers from 35 through 85 form an arithmetic sequence with first term 35 and each subsequent term 2 more than the preceding term. Thus the sum 35 + 37 + 39 + … + 85 can be found as follows:

1st term	35	= 35		
2nd term	37	= 35	+ 1(2)	
3rd term	39	= 35	+ 2(2)	
4th term	41	= 35	+ 3(2)	
…	…	… …	… …	
26th term	85	= 35	+ 25(2)	

$$\begin{aligned} \text{Sum} &= 35(26) + (1 + 2 + 3 + \ldots + 25)(2) \\ &= 35(26) + \frac{(25)(26)}{2}(2) \\ &\qquad \text{see note below} \\ &= 910 + 650 \\ &= 1{,}560 \end{aligned}$$

Note that if $s = 1 + 2 + 3 + \ldots + 25$, then $2s = (1 + 2 + 3 + \ldots + 25) + (25 + 24 + 23 + \ldots + 1)$, and so $2s = (1 + 25) + (2 + 24) + (3 + 23) + \ldots + (25 + 1) = (25)(26)$. Therefore, $s = \frac{(25)(26)}{2}$.

Alternatively, to determine the number of odd integers from 35 to 85, inclusive, consider that 3 of them (35, 37, and 39) have tens digit 3. Half of the integers with tens digit 4 are odd, so 5 of the odd integers between 35 and 85, inclusive, have tens digit 4. Similarly, 5 of the odd integers between 35 and 85, inclusive, have tens digit 5; 5 have tens digit 6; and 5 have tens digit 7. Finally, 3 have tens digit 8 (81, 83, and 85), and so the number of odd integers between 35 and 85, inclusive, is 3 + 5 + 5 + 5 + 5 + 3 = 26. Now, let $S = 35 + 37 + 39 + \ldots + 85$. Then, $S = 85 + 83 + 81 + \ldots + 35$, and it follows that
$2S = (35 + 85) + (37 + 83) + (39 + 81) + \ldots + (85 + 35) = (120)(26)$. Thus,
$S = 35 + 37 + 39 + \ldots + 85 = \frac{(120)(26)}{2} = 1{,}560$.

The correct answer is A.

•

For all numbers a, b, c, and d, $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is defined by the equation $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$. Which of the following is

. $|s \quad t|$ $|-t \quad 2|$ $|2 \quad 2|$

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

equal to $\begin{vmatrix} s & t \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} -t & 2 \\ s & 4 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ t & s \end{vmatrix}$?

A. $\begin{vmatrix} s & t \\ 1 & 5 \end{vmatrix}$

B. $\begin{vmatrix} s & t \\ 7 & 1 \end{vmatrix}$

C. $\begin{vmatrix} s & t \\ 5 & 7 \end{vmatrix}$

D. $\begin{vmatrix} s & -t \\ 1 & 5 \end{vmatrix}$

E. $\begin{vmatrix} s & -t \\ 1 & 7 \end{vmatrix}$

Algebra Formulas

First, expand the given expression using the given definition.

$$\begin{vmatrix} s & t \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} -t & 2 \\ s & 4 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ t & s \end{vmatrix} = (3s - t) - (-4t - 2s) + (2s - 2t) = 3s - t + 4t + 2s + 2s - 2t = 7s + t$$

Next, compare the result, $7s + t$, with the expanded versions of the answer choices.

$$\text{A} \begin{vmatrix} s & t \\ 1 & 5 \end{vmatrix} = 5s - t \quad \text{(not correct)}$$

$$\text{B} \begin{vmatrix} s & t \\ 7 & 1 \end{vmatrix} = s - 7t \quad \text{(not correct)}$$

$$\text{C} \begin{vmatrix} s & t \\ 5 & 7 \end{vmatrix} = 7s - 5t \quad \text{(not correct)}$$

$$\text{D} \begin{vmatrix} s & -t \\ 1 & 5 \end{vmatrix} = 5s + t \quad \text{(not correct)}$$

$$\text{E} \begin{vmatrix} s & -t \\ 1 & 7 \end{vmatrix} = 7s + t \quad \text{(correct)}$$

The correct answer is E.

•

In a certain sequence, each term after the first term is one-half the previous term. If the tenth term of the sequence is between 0.0001 and 0.001, then the twelfth term of the sequence is between

A. 0.0025 and 0.025

B. 0.00025 and 0.0025

C. 0.000025 and 0.00025

D. 0.0000025 and 0.000025

E. 0.00000025 and 0.0000025

Arithmetic Sequences

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Let a_n represent the nth term of the sequence. It is given that each term after the first term is $\frac{1}{2}$ the previous term and that $0.0001 < a_{10} < 0.001$. Then for a_{11}, $\frac{0.0001}{2} < a_{11} < \frac{0.001}{2}$, or $0.00005 < a_{11} < 0.0005$. For a_{12}, $\frac{0.00005}{2} < a_{12} < \frac{0.0005}{2}$, or $0.000025 < a_{12} < 0.00025$. Thus, the twelfth term of the sequence is between 0.000025 and 0.00025.

The correct answer is C.

•

A certain drive-in movie theater has a total of 17 rows of parking spaces. There are 20 parking spaces in the first row and 21 parking spaces in the second row. In each subsequent row there are 2 more parking spaces than in the previous row. What is the total number of parking spaces in the movie theater?

A. 412

B. 544

C. 596

D. 632

E. 692

Arithmetic Operations on Integers

Row	Number of parking spaces
1st row	20
2nd row	21
3rd row	21 + 1(2)
4th row	21 + 2(2)
…	… … …
17th row	21 + 15(2)

Then, letting S represent the total number of parking spaces in the theater,

$$\begin{aligned} S &= 20 + (16)(21) + (1+2+3+\ldots+15)(2) \\ &= 20 + 336 + \frac{(15)(16)}{2}(2) \text{ see note below} \\ &= 356 + 240 \\ &= 596 \end{aligned}$$

Note that if $s = 1+2+3+\ldots+15$, then $2s = (1+2+3+\ldots+15) + (15+14+13+\ldots+1)$, and so $2s = (1+15)+(2+14)+(3+13)+\ldots+(15+1) = (15)(16)$. Therefore, $s = \frac{(15)(16)}{2}$.

The correct answer is C.

•

Ada and Paul received their scores on three tests. On the first test, Ada's score was 10 points higher than Paul's score. On the second test, Ada's score was 4 points higher than Paul's score. If Paul's average (arithmetic mean) score on

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

On the second test, Ada's score was 4 points higher than Paul's score. If Paul's average (arithmetic mean) score on the three tests was 3 points higher than Ada's average score on the three tests, then Paul's score on the third test was how many points higher than Ada's score?

A. 9

B. 14

C. 17

D. 23

E. 25

Algebra Statistics

Let a_1, a_2, and a_3 be Ada's scores on the first, second, and third tests, respectively, and let p_1, p_2, and p_3 be Paul's scores on the first, second, and third tests, respectively. Then, Ada's average score is $\frac{a_1 + a_2 + a_3}{3}$ and Paul's average score is $\frac{p_1 + p_2 + p_3}{3}$. But Paul's average score is 3 points higher than Ada's average score, so $\frac{p_1 + p_2 + p_3}{3} = \frac{a_1 + a_2 + a_3}{3} + 3$. Also, it is given that $a_1 = p_1 + 10$ and $a_2 = p_2 + 4$, so by substitution, $\frac{p_1 + p_2 + p_3}{3} = \frac{(p_1 + 10) + (p_2 + 4) + a_3}{3} + 3$. Then $p_1 + p_2 + p_3 = (p_1 + 10) + (p_2 + 4) + a_3 + 9$ and so $p_3 = a_3 + 23$. On the third test, Paul's score was 23 points higher than Ada's score.

The correct answer is D.

•

The price of a certain stock increased by 0.25 of 1 percent on a certain day. By what fraction did the price of the stock increase that day?

A. $\frac{1}{2{,}500}$

B. $\frac{1}{400}$

C. $\frac{1}{40}$

D. $\frac{1}{25}$

E. $\frac{1}{4}$

Arithmetic Percents

It is given that the price of a certain stock increased by 0.25 of 1 percent on a certain day. This is equivalent to an increase of $\frac{1}{4}$ of $\frac{1}{100}$, which is $\left(\frac{1}{4}\right)\left(\frac{1}{100}\right)$, and $\left(\frac{1}{4}\right)\left(\frac{1}{100}\right) = \frac{1}{400}$.

The correct answer is B.

•

For each trip, a taxicab company charges \$4.25 for the first mile and \$2.65 for each additional mile or fraction thereof. If the total charge for a certain trip was \$62.55, how many miles at most was the trip?

A. 21

B. 22

C. 23

D. 24

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

E. 25

Arithmetic Applied Problems

Subtracting the charge for the first mile leaves a charge of \$62.55 – \$4.25 = \$58.30 for the miles after the first mile. Divide this amount by \$2.65 to find the number of miles to which \$58.30 corresponds: $\frac{58.30}{2.65} = 22$ miles.

Therefore, the total number of miles is at most 1 (the first mile) added to 22 (the number of miles after the first mile), which equals 23.

The correct answer is C.

•

When 24 is divided by the positive integer n, the remainder is 4. Which of the following statements about n must be true?

I. n is even.

II. n is a multiple of 5.

III. n is a factor of 20.

A. III only

B. I and II only

C. I and III only

D. II and III only

E. I, II, and III

Arithmetic Properties of Numbers

Since the remainder is 4 when 24 is divided by the positive integer n and the remainder must be less than the divisor, it follows that $24 = qn + 4$ for some positive integer q and $4 < n$, or $qn = 20$ and $n > 4$. It follows that $n = 5$, or $n = 10$, or $n = 20$ since these are the only factors of 20 that exceed 4.

I. n is not necessarily even. For example, n could be 5.

II. n is necessarily a multiple of 5 since the value of n is either 5, 10, or 20.

III. n is a factor of 20 since $20 = qn$ for some positive integer q.

The correct answer is D.

•

Terry needs to purchase some pipe for a plumbing job that requires pipes with lengths of 1 ft 4 in, 2 ft 8 in, 3 ft 4 in, 3 ft 8 in, 4 ft 8 in, 5 ft 8 in, and 9 ft 4 in. The store from which Terry will purchase the pipe sells pipe only in 10-ft lengths. If each 10-ft length can be cut into shorter pieces, what is the minimum number of 10-ft pipe lengths that Terry needs to purchase for the plumbing job?

(Note: 1 ft = 12 in)

A. 3

B. 4

C. 5

D. 6

E. 7

Arithmetic Operations with Integers; Measurement Conversion

The 7 lengths of pipe Terry needs total 30 feet plus 8 inches, which means Terry will need to buy at least 4 pipes, each 10 feet long. Four pipes will suffice if Terry cuts pieces of the following lengths:

1st pipe: 9 feet 4 inches (with 8 inches left)

2nd pipe: 5 feet 8 inches and 3 feet 8 inches (with 8 inches left)

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

3rd pipe: 4 feet 8 inches, 3 feet 4 inches, and 1 foot 4 inches (with 8 inches left)

4th pipe: 2 feet 8 inches (with 7 feet 4 inches left)

The correct answer is B.

•

What is the thousandths digit in the decimal equivalent of $\frac{53}{5{,}000}$?

A. 0

B. 1

C. 3

D. 5

E. 6

Arithmetic Place Value

$\frac{53}{5{,}000} = \frac{106}{10{,}000} = 0.0106$ and the thousandths digit is 0.

The correct answer is A.

•

If a equals the sum of the even integers from 2 to 20, inclusive, and b equals the sum of the odd integers from 1 to 19, inclusive, what is the value of $a - b$?

A. 1

B. 10

C. 19

D. 20

E. 21

Arithmetic Series and Sequences

$$\begin{aligned} a &= 2 + 4 + 6 + \cdots + 20 \\ b &= 1 + 3 + 5 + \cdots + 19 \\ a - b &= (2-1) + (4-3) + (6-5) + \cdots + (20-19) \\ &= 1 + 1 + 1 + \cdots + 1 \end{aligned}$$

Since there are 10 even numbers in the expression for a above—note that the expression for a can be written as 2(1 + 2 + 3 + … + 10)—it follows that there are 10 occurrences of 1 in the expression for $a - b$, and hence $a - b = 10 \times 1 = 10$.

The correct answer is B.

•

If a, b, c, and d are consecutive even integers and $a < b < c < d$, then $a + b$ is how much less than $c + d$?

A. 2

B. 4

C. 6

D. 8

E. 10

Algebra Series and Sequences; Simplifying Algebraic Expressions

Since a, b, c, and d are consecutive even integers in this order, we have $b = a + 2$, $c = b + 2 = a + 4$, and

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

Since a, b, c, and d are consecutive even integers in this order, we have $b = a + 2$, $c = b + 2 = a + 4$, and $d = c + 2 = a + 6$. Therefore, the amount by which $a + b$ is less than $c + d$ is $(c + d) - (a + b) = (a + 4 + a + 6) - (a + a + 2) = 8$.

The correct answer is D.

•

A retailer sold an appliance for $80. If the retailer's gross profit on the appliance was 25 percent of the retailer's cost for the appliance, how many dollars was the retailer's gross profit?

A. $10

B. $16

C. $20

D. $24

E. $25

Algebra Applied Problems; First-Degree Equations

Let $\$C$ be the cost of the appliance. The appliance sold for $80 and the profit was 25% of the cost. Since selling price equals cost plus profit, it follows that $80 = C + 0.25C = 1.25C$. Dividing both sides by 1.25 (equivalently, dividing both sides by $\frac{5}{4}$), we have $C = 64$. Therefore, the profit is $0.25(\$C) = \frac{1}{4} \times \$64 = \$16$.

The correct answer is B.

•

Beth has a collection of 8 boxes of clothing for a charity, and the average (arithmetic mean) number of pieces of clothing per box is c. If she replaces a box in the collection that contains 12 pieces of clothing with a box that contains 22 pieces of clothing, what is the average number of pieces of clothing per box for the new collection, in terms of c?

A. $c - \frac{5}{4}$

B. $c + \frac{5}{4}$

C. $8 - \frac{10}{c}$

D. $8 + \frac{10}{c}$

E. $8c - 10$

Algebra Average

Before the replacement, the total number of pieces of clothing in the 8 boxes was $8c$. After the replacement, the total number of pieces of clothing in the 8 boxes was $8c - 12 + 22 = 8c + 10$. Therefore, the average number of pieces of clothing per box for the new collection is $\frac{8c + 10}{8} = \frac{8c}{8} + \frac{10}{8} = c + \frac{5}{4}$.

The correct answer is B.

$$\frac{1 + 0.0001}{0.04 + 10}$$

•

The value of the expression above is closest to which of the following?

A. 0.0001

B. 0.001

C. 0.01

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

D. 1

E. 10

Arithmetic Estimation

$\frac{1+0.0001}{0.04+10} = \frac{1.0001}{10.04} \approx \frac{1}{10} = 0.1$ Alternatively (and less heuristically), because a small relative change (i.e., a small percentage change) in the value of the numerator of a fraction produces a small relative change in the value of the fraction, and similarly for the denominator, it follows that $\frac{1+0.0001}{0.04+10} \approx \frac{1}{0.04+10} \approx \frac{1}{10} = 0.1$

The correct answer is C.

•

If $x+1=t$ and $t=3-x$, then $x=$

A. –2

B. –1

C. 0

D. 1

E. 2

Algebra First-Degree Equations

$x+1 = 3-x$	substitution
$x+x = 3-1$	add $x-1$ to both sides
$2x = 2$	combine like terms
$x = 1$	divide both sides by 2

The correct answer is D.

•

If $x=kc$ and $y=kt$, then $y-x=$

A. $k(t-c)$

B. $k(c-t)$

C. $c(k-t)$

D. $t(k-c)$

E. $k(1-t)$

Algebra Simplifying Algebraic Expressions

$y-x=kt-kc$	given
$y-x=k(t-c)$	distributive property

The correct answer is A.

•

If k is a positive even integer, which of the following must be an odd integer?

I. k^2-3k+4

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

II. $k^5 + 3$

III. $7k - 7$

A. II only

B. III only

C. I and III only

D. II and III only

E. I, II, and III

Arithmetic Properties of Integers

The table below shows some results for adding, subtracting, and multiplying even and odd integers. For example, an even integer plus an even integer is an even integer (for example, 4 + 8 equals 12, which is even) and an odd integer times an odd integer is an odd integer (for example, 3 × 7 equals 21, which is odd).

even	±	even	is	even
even	±	odd	is	odd
odd	±	odd	is	even
even	×	even	is	even
even	×	odd	is	even
odd	×	odd	is	odd

Now apply these results to the expressions in I, II, and III when k is an even integer.

I. $k^2 - 3k + 4$ is even − even + even, which is **EVEN**.

II. $k^5 + 3$ is even + odd, which is **ODD**.

III. $7k - 7$ is even – odd, which is **ODD**.

The correct answer is D.

Questions 83 to 151 — Difficulty: Medium

83. If $\frac{1}{2}$ the result obtained when 2 is subtracted from $5x$ is equal to the sum of 10 and $3x$, what is the value of x?

A. –22

B. –4

C. 4

D. 18

E. 22

Algebra First-Degree Equations

The result obtained when 2 is subtracted from $5x$ is $5x - 2$, and the sum of 10 and $3x$ is $10 + 3x$.

Therefore, it is given that $\frac{1}{2}$ of $5x - 2$ is equal to $10 + 3x$, or $\frac{1}{2}(5x - 2) = 10 + 3x$.

$$\frac{1}{2}(5x - 2) = 10 + 3x \quad \text{given}$$

$$5x - 2 = 20 + 6x \quad \text{multiply both sides by 2}$$

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

$5x = 2 \quad 20 + 6x$ multiply both sides by 2

$-22 = x$ subtract both $5x$ and 20 from both sides

The correct answer is A.

84. If Car A took n hours to travel 2 miles and Car B took m hours to travel 3 miles, which of the following expresses the time it would take Car C, traveling at the average (arithmetic mean) of those rates, to travel 5 miles?

A. $\dfrac{10nm}{3n+2m}$

B. $\dfrac{3n+2m}{10(n+m)}$

C. $\dfrac{2n+3m}{5nm}$

D. $\dfrac{10(n+m)}{2n+3m}$

E. $\dfrac{5(n+m)}{2n+3m}$

Algebra Applied Problems

This is a rate problem that can be solved by several applications of the formula

rate × time = distance.

Let r_A and r_B be the rates, respectively and in miles per hour, of Car A and Car B. Then omitting units for simplicity, for Car A this formula becomes $r_A \times n = 2$, or $r_A = \dfrac{2}{n}$, and for Car B this formula becomes $r_B \times m = 3$, or $r_B = \dfrac{3}{m}$. Thus, the average of the two rates is $\dfrac{1}{2}(r_A + r_B) = \dfrac{1}{2}\left(\dfrac{2}{n} + \dfrac{3}{m}\right) = \dfrac{1}{2}\left(\dfrac{2m+3n}{mn}\right) = \dfrac{3n+2m}{2mn}$. Therefore, if t is the desired time, in hours, that Car C traveled, then the above formula for Car C becomes $\dfrac{3n+2m}{2mn} \times t = 5$, or $t = 5 \times \dfrac{2mn}{3n+2m} = \dfrac{10mn}{3n+2m}$.

The correct answer is A.

85. If x, y, and k are positive and x is less than y, then $\dfrac{x+k}{y+k}$ is

A. 1

B. greater than $\dfrac{x}{y}$

C. equal to $\dfrac{x}{y}$

D. less than $\dfrac{x}{y}$

E. less than $\dfrac{x}{y}$ or greater than $\dfrac{x}{y}$, depending on the value of k

Algebra Ratios

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

$x < y$	given
$kx < ky$	multiply by positive k
$xy + kx < xy + ky$	add xy
$x(y+k) < y(x+k)$	factor
$x < \dfrac{y(x+k)}{y+k}$	divide by positive $y+k$
$\dfrac{x}{y} < \dfrac{x+k}{y+k}$	divide by positive y

Thus, $\dfrac{x+k}{y+k} > \dfrac{x}{y}$.

The correct answer is B.

86. Consider the following set of inequalities: $p > q$, $s > r$, $q > t$, $s > p$, and $r > q$. Between which two quantities is no relationship established?

A. p and r

B. s and t

C. s and q

D. p and t

E. r and t

Algebra Order

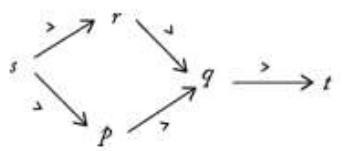
Using $r > q$ and $q > t$ gives $r > t$, so a relationship is established between r and t. The correct answer is NOT E.

Using $p > q$ and $q > t$ gives $p > t$, so a relationship is established between p and t. The correct answer is NOT D.

Using $s > r$ and $r > q$ gives $s > q$, so a relationship is established between s and q. The correct answer is NOT C.

Using $s > r$, $r > q$, and $q > t$ gives $s > t$, so a relationship is established between s and t. The correct answer is NOT B.

Alternately, the diagram below shows the given relationships and does not establish a relationship between p and r.



The correct answer is A.

87. Carl averaged $2m$ miles per hour on a trip that took him h hours. If Ruth made the same trip in $\dfrac{2}{3}h$ hours, what was her average speed in miles per hour?

A. $\dfrac{1}{3}mh$

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. $\frac{2}{3}mh$

C. m

D. $\frac{3}{2}m$

E. $3m$

Algebra Applied Problems

Using distance = rate × time, the distance Carl traveled on the trip was $2mh$ miles. Using $\text{rate} = \frac{\text{distance}}{\text{time}}$, Ruth's rate was $\frac{2mh}{\frac{2}{3}h} = \frac{3}{2}(2m) = 3m$.

The correct answer is E.

88. Of three persons, two take relish, two take pepper, and two take salt. The one who takes no salt takes no pepper, and the one who takes no pepper takes no relish. Which of the following statements must be true?

I. The person who takes no salt also takes no relish.

II. Any of the three persons who takes pepper also takes relish and salt.

III. The person who takes no relish is not one of those who takes salt.

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

Arithmetic Sets (Venn Diagrams)

Although this problem can be solved by the use of a Venn diagram, it is probably simpler to use ordinary reasoning. The single person who takes no salt takes no pepper, and the single person who takes no pepper takes no relish, so exactly one person does not take any of the three. Thus, each of the other two people take all three. The table below shows these results where Person 1 does not take any of the three and Persons 2 and 3 each take all three.

Person	Relish	Pepper	Salt
1	no	no	no
2	yes	yes	yes
3	yes	yes	yes

The only person who takes no salt is Person 1, who also takes no relish, so I must be true.

The only people who take pepper are Persons 2 and 3, and each of them also takes relish and salt, so II must be true.

The only person who takes no relish is Person 1, who is not a person who takes salt, so III must be true.

The correct answer is E.

89. If the smaller of 2 consecutive odd integers is a multiple of 5, which of the following could NOT be the sum of these 2 integers?

A. –8

DO NOT COPY filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

B. 12

C. 22

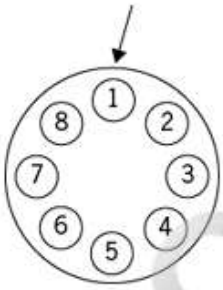
D. 52

E. 252

Algebra Operations with Integers

Since the smaller of the 2 consecutive odd integers is a multiple of 5, let it be represented by $5n$ for some integer n. Then the other odd integer can be represented by $5n + 2$. The sum of these two integers is $10n + 2$. The sum is –8 when $n = -1$ and $5n = (5)(-1)$ is odd. The sum is 12 when $n = 1$ and $5n = (5)(1)$ is odd. The sum is 22 when $n = 2$, but $5n = (5)(2)$ is not odd. There is no need to check the D and E because it has been determined that 22 cannot be the sum of the 2 consecutive odd integers. For completeness, the sum is 52 when $n = 5$ and $5n = (5)(5)$ is odd. The sum is 252 when $n = 25$ and $5n = (5)(25)$ is odd.

The correct answer is C.



90. Eight light bulbs numbered 1 through 8 are arranged in a circle as shown above. The bulbs are wired so that every third bulb, counting in a clockwise direction, flashes until all bulbs have flashed once. If the bulb numbered 1 flashes first, which numbered bulb will flash last?

A. 2

B. 3

C. 4

D. 6

E. 7

Arithmetic Properties of Integers

The easiest way to do this problem might be by just counting every third bulb going clockwise around the circle starting at Bulb 1, which flashes, skipping 2 bulbs and getting to Bulb 4, which flashes, skipping 2 bulbs and getting to Bulb 7, which flashes, skipping 2 bulbs and getting to Bulb 2, which flashes, skipping 2 bulbs and getting to Bulb 5, which flashes, skipping 2 bulbs and getting to Bulb 8, which flashes, skipping 2 bulbs and getting to Bulb 3, which flashes, and finally skipping 2 bulbs and getting to Bulb 6, which flashes. Now, all 8 bulbs have flashed once and the last one to flash was Bulb 6.

The correct answer is D.

Closing Prices of Stock X
During a Certain Week
(in dollars)

Monday	Tuesday	Wednesday	Thursday	Friday
21	19	22	$24\frac{1}{2}$	23

91. A certain financial analyst defines the "volatility" of a stock during a given week to be the result of the following procedure: find the absolute value of the difference in the stock's closing price for each pair of consecutive days

DO NOT COPY
filippo.fiz@uptoten.it

Printed by: filippo.fiz@uptoten.it. Printing is for personal, private use only. No part of this book may be reproduced or transmitted without publisher's prior permission. Violators will be prosecuted.

in the week and then find the average (arithmetic mean) of these 4 values. What is the volatility of Stock X during the week shown in the table?

A. 0.50

B. 1.80

C. 2.00

D. 2.25

E. 2.50

Arithmetic Statistics

The volatility of Stock X during the week is the average of the 4 values associated with the 4 pairs of consecutive days during the week.

DO NOT COPY
filippo.fiz@uptoten.it